Research statement

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November 2026, yspennstate.github.io, code and results at github.com/yspennstate/math-brain-public and github.com/yspennstate/brain-framework-public

I work in probability, random matrices and the mathematics of deep learning. I did my PhD at Penn State with Leonid Berlyand, and I am now a postdoctoral fellow at the Hebrew University of Jerusalem, hosted by Sasha Sodin and Amit Daniely.

Over the past year I have directed AI agents in proving theorems, on open problems and on computational problems, and I have built the memory they work from. I call it the brain. It is a graph whose nodes are mathematical results, dead proof routes, recorded mistakes, code and tools, and whose edges say how these are related and how often they get used together. The work so far comes to more than a million Claude model calls and more than 17,000 Codex sessions, about half a trillion input tokens and more than 1.5 billion output tokens. The overall goal is agency, meaning agents that carry a hard problem further before a person has to step in. Recall, finding the right node when an agent needs it, helps with that, but it is only the first layer of what I have built. The brain is also a mathematical object, built by agents, and I study it with what I was trained on (random walks, spectral theory, random matrices).

1. Why agents need a memory

Often when agents do research they get stuck in a loop where they keep doing the same thing over and over again. An agent proves a lemma it already proved last week or tries a route the agent next to it just gave up on, and when its working context fills up and is wiped it can lose a theorem. A research group does not work that way, because people remember what was tried and why it failed. I want agents to have that memory and to reach for it at the right moment. So I count how far an agent gets on a fixed budget and how often it repeats a step already on record (and what a right answer costs).

2. What the brain is

There are two stores, one inside the other. The inner one, the math brain, records every item with its type, and a snapshot of it is public [1]. Its items, which I call cards, are definitions, theorems, lemmas, propositions, proof techniques, worked proofs, counterexamples, verified computations, open problems and dead routes (a route is a proof strategy written as a sequence of steps, and a dead one carries the step at which it failed). Each card carries its sources and a status, which is verified, provisional or superseded. The links are the relations a mathematician uses, such as depends on, proves, refutes, generalizes and cites. A wrong card is never deleted. A later card supersedes it, so the next agent sees the error and the fix. The working copy held 90,596 cards on 28 October 2026, and the public snapshot of 1 November has 13,344 cards and 23,518 typed links.

The research brain is the larger private store around it, and its machinery is public [23]. It holds more than 30,000 written memories (results, rules, mistakes with their mechanism), together with the cards, prompts, tools, files, keywords and more than 360 reflexes (a reflex is a recorded mistake compiled into a trigger and a rule). Agents write some of its edges when they connect two results, and the rest build up by themselves. When agents use two nodes in one action, or one right after the other, the edge between them gets stronger, and when they stop it fades. So agents use a theorem, a proposition or a tool a few times and all of a sudden there is a connection. That itself teaches agents what is connected, what pieces of code, what keywords, when a theorem and a piece of code go together. On 1 November my main machine held 349,065 nodes and 15.5 million learned edges, built from 15.3 million recorded events over 5,747 sessions. Some of these edges join things I had not connected myself (section 6 proposes a test).

The brain is not just an index. It is an index together with mechanisms that improve an agent's ability to push proofs forward and do research without getting stuck in loops. The network sits in the background like a subconscious, and the hard part is getting what it holds into the agent's context when it is needed. Recall is what an agent asks for, but an agent cannot search for what it does not know it needs. So a layer of hooks (small programs that run on every prompt and tool call) puts the nearest cards, and the mistakes

recorded in that territory, in front of the agent unasked. Agents also write new hooks, skills and workflows (written procedures another agent can load and follow) as the brain develops. So the entire thing is a growth mechanism where agents can work faster, recall better and push things forward while the work itself adds to what they recall from. The store was built the same way. Thousands of agent sessions, directed by me, read the literature of probability and deep learning, wrote the cards, proved and refuted what they could, and connected results across fields.

3. The mathematics of the brain

Recall as I built it can run as diffusion on a graph. Write the store as $G = (V, E, w)$, with cards V, links E and weights $w(i, j) > 0$. Let A be the adjacency matrix, D the diagonal matrix of row sums and $P = D^{-1}A$, so that row i of P says what fraction of card i's weight goes to each neighbour. A question q gives each card two scores, the BM25 score $\ell_i(q)$ of its words (the standard word-match score [2], in which rare words count for more) and the cosine $m_i(q)$ between embedding vectors, which is high when two texts say the same thing in different words. After standardizing each over the candidate cards, the seed on the research brain is $s = 0.7\, z(\ell) + 0.3\, z(m)$, with $s = 0$ off the candidates (the math brain adds scores for symbols and formulas), and activation can then spread along the links for at most two hops, with fixed weights β_1, β_2:

$$a(q) = s(q) + \beta_1 P^{\mathsf{T}} s(q) + \beta_2\, (P^{\mathsf{T}})^2 s(q).$$

The entry $(P^{\mathsf{T}}s)_i = \sum_j P_{ji} s_j$ is what card i collects from the cards that link to it, so a card the question's words never touch can still enter the pool if well-matched cards point at it. When $\beta_k = c^k$ these are, up to the factor $1 - c$, the first terms of the personalized PageRank series $\sum_k c^k (P^{\mathsf{T}})^k s$ [3, 4]. That is the model. The code uses A^{T} in place of P^{T}, with 0.55 per link, because dividing by the row sums changed first place on only one question of 381. Undivided, a card passes its whole score to every neighbour, and many steps would concentrate the mass on the leading eigenvector (eigenvector centrality), whatever the question was. Two steps hurt us in a different way. Sixteen junk cards, 1.4 percent of an early store, were not hubs (degree 3 to 5), but 92 percent of their edges stayed inside the group. They kept feeding each other and held the top three places on 39 percent of the questions the store could answer. We took the junk out of the seeds and turned the walk off.

On my stores so far the walk mostly reorders what the words already found. With a synthetic new memory on the research brain's live graph and real prompts, spreading from the ten best word matches never put it in the top ten, in 7,440 trials at each of 1 to 20 edges. The links got it into the pool and no further. On real cards the walk probably adds little because linked cards have about ten times the word overlap of random pairs when rare words count for more. Recall now runs without spreading by default and the agent walks the links by hand when it wants to (very cheap, see section 4). I doubt this is general. DAPS [5] reports strong recall on held-out Mathlib theorems with an encoder over typed dependencies, and I want a theorem that says when links help (section 5).

The learned edges of section 2 form a second graph, the use graph. A node's weight is its base-level activation in the sense of ACT-R, a standard model of human memory [6, 7], and an edge's weight is a count of uses together, read with a decay:

$$B_i(t) = \ln \sum_k (t - t_{ik})^{-d}, \qquad w_{ij}(t) = \sum_k e^{-(t - t_{ijk})/\tau},$$

where t_{ik} are the past times node i was used and t_{ijk} the past times i and j were used together, and d and τ set how fast each fades. With $d = \frac{1}{2}$ and time in days, one use yesterday contributes 1 and ten uses a year ago contribute $10/\sqrt{365} \approx 0.52$, so the memory touched yesterday outranks the one that was popular last year.

The hooks of section 2 also have to know when to stay quiet. If a hook writes a note on every prompt the agents learn to ignore it. Let $a_{(1)} \geq a_{(2)}$ be the two largest scores for a prompt, and serve a note only if $a_{(1)} \geq \theta$ and $a_{(1)} - a_{(2)} \geq \gamma$. A threshold on the word score alone already separates 2,000 questions the store can answer from 60 it cannot, with AUC 0.997 and 6.4 percent of the answerable ones wrongly getting no note [1]. New cards have to pass a test too. Let $\mathrm{acc}(S)$ be the fraction of a held-out bank of questions

whose answer card ranks in the first eight when recall runs over the store S. A packet Π of new cards enters only if $\mathrm{acc}(S \cup \Pi) \geq \mathrm{acc}(S)$ on a copy, so growth cannot lower the score on that bank.

To stop the loops of section 1 we built the frontier layer. Each open problem agents are working on has a file with the statement, the latest papers and the step where their proofs fail, what is settled, which routes are dead and why, and the next targets in order. An agent starts from that failing step and checks its route against the settled facts and the dead routes. If it repeats a dead route, the agent is stopped unless it writes a concrete reason its route differs. On 28 October the layer covered 64 problems and 2,623 routes. In a preregistered test with a blind judge on two open problems, eight comparisons in all, a plan written with a digest of what the store held on that problem (the digest's second version) was preferred over a plan written without it 7 of 8 times and had 8 redundant or dead-end steps against 17 [1]. That test judged plans and not proofs, and whether the layer gets further on hard problems is the main test of next year.

Speed matters too, because agents stop calling a memory that is slow. When agents measured recall in October, the gains came from not starting over on each call and none came from ranking. A resident server took a recall from 26.8 to 7.4 seconds at the median, with identical answers [1].

4. The tests

Every test compares setups (arms) on the same questions with the same model and time cap, and the p values are exact binomial tests on the pairs where exactly one arm was right. The table's first test has 988 items, lookups over the store's records, with answer keys made by script, and proof audits, where an agent names a proof's first bad step or passes it. There the memory arm had recall and the hooks and no file tools, the repositories arm could read, list and search the research repositories that hold the same records, and the bare arm had the model alone. In the second, the git test, both arms had the repositories and one of them also had recall. The public repository has the design and aggregates of every test, including those that showed no gain [1].

Test	Agents	Other arm	Memory	Discordant, p
988 items from a bank of 923 scripted lookups and 77 proof audits, 600 s each	Claude	82.0%	95.4%	146 to 13, 1.2e−29
2,000 questions on files in 113 git repositories, one recall a question (both arms reached 1,726)	Codex	91.4%	93.7%	83 to 43, 4.6e−4
160 proof audits with planted faults, the brain's notes against the same kind of notes scrambled, replicated on a second bank		117	128	18 to 7, 0.043

On the 988 items the bare arm got 18.9 percent. The memory arm took about half the time of the repositories arm (120 seconds an item against 211) and timed out a fifth as often (3.7 percent against 17.5). Its finished sessions read 20,000 tokens against 80,000, and a right answer cost $0.074 against $0.153 (counting a second pass with a longer limit). On the 806 items both arms finished they were tied, 99.3 and 99.4 percent (a set selected by finishing). So the gain is more questions finished on the same budget, which was the whole idea in section 1. The git test is harder for a memory, because both arms have the files and the questions are about those files. Where both arms committed to an answer they were right about equally often. The lead comes from the repositories arm giving up more often. The memory arm was slower, by about one recall on a busy machine (a median of 67 seconds against 13). The git scores come from a regrade done after the answers were known, under three rules applied to both arms.

The links and the notes did not raise the score, but they cut the cost. On 100 questions on results in the store, agents that also got the hook's notes and could step from card to card along the links scored 82 against 83 for recall alone, on 7.6 times fewer input tokens (1.8 times fewer uncached) and 78 session-minutes against 666 [1].

5. Results, and the theory I want

What I care about in the end is the mathematics. AI systems did much of the mathematics in the following results, I directed the work and I am responsible for it, and each paper has a declaration of who did what.

They have been audited by independent agent runs and are not yet refereed. The agents worked from this memory, and I do not claim the memory caused any proof.

Ross [8] conjectured, in Leskelä's formulation [9], that a queue fed by a randomly varying arrival rate has no more mean workload when the environment moves faster. Take a rate-one single-server queue with Poisson arrivals of intensity $\varepsilon F(ct)$ and let $w_\varepsilon(c)$ be its stationary mean workload. The conjecture says w_ε is nonincreasing in c for every stationary ergodic F, and it is false. An explicit 64-state Markov intensity, with every transition rate positive and service times uniform on $[0.99, 1.01]$, has $w_\varepsilon(3\pi) - w_\varepsilon(2\pi) > \varepsilon^2/600$ for $0 < \varepsilon \le 10^{-4}$. The gap is proved only in light traffic and it is very small there. The second-order workload coefficient is nonincreasing in every bounded environment exactly when $s \mapsto \mathbb{E}(1 - \cos sB)/s^2$ is nonincreasing, for service time B [10].

In planar percolation, let f be the crossing function of a square of side R, $\mathcal{P}$ its pivotal set and $\mathcal{S}$ its spectral sample, the random set with law $\hat{f}(A)^2$. At the uniform measure the two sets have the same one- and two-point inclusion probabilities. For critical bond percolation on the square lattice and site percolation on the triangular lattice, the Shannon entropies $H(\mathcal{S})$ and $H(\mathcal{P})$ are both comparable to the total influence, which is of order $R^2\alpha_4(R)$ with α_4 the four-arm probability. The spectral bound answers the square-crossing question of Garban, Pete and Schramm [11, Section 11, Question 1], and the pivotal bound gives the other square-crossing bound in Kalai's Conjecture 8 [12]. It is a percolation case of the Fourier entropy-influence conjecture [13] and says nothing about general Boolean functions. Equal inclusion statistics and comparable entropy still do not make the sets alike. For critical triangular-site crossings the two laws are asymptotically mutually singular, which answers the discrete case of their Question 5 [11]: an elementary triangular face never lies in a pivotal set and appears with probability tending to one in the spectral sample, and the overlap on nonempty sets decays as $R^{-11/12+o(1)}$ [14]. This does not settle the continuum limits or the square-lattice bond case.

Vu conjectured [15, Conjecture 9.5] that an $n \times n$ random sign matrix M_n has all its eigenvalues in $\mathbb{Z}[i]$ with probability $2^{-(1+o(1))n^2}$. An explicit family gives $\Pr(\mathrm{Spec}(M_n) \subset \mathbb{Z}[i]) \ge 3^{\lfloor n/2 \rfloor} 2^{-\lfloor n^2/2 \rfloor}$, about the square root of that, so the conjecture is false [16]. Lyons and Peres [17, Question 16.7] asked whether simple random walks on the base-2 and base-3 coding trees of the Cantor set give mutually singular harmonic measures on it, and they do [18]. For Aldous and Fill's Open Problem 7.30 [19], on every finite connected 1-walk-regular graph the heat-kernel ratio $H_t(x,y)/H_t(x,x)$, with $H_t = e^{-t(I-A/d)}$ on a d-regular graph, is strictly increasing in t across every edge [20]. In an excited random walk on the integers, the i-th visit to a site uses the i-th cookie of that site's stack, which biases the step. For IID stacks with nonnegative drift under weak ellipticity (Kosygina and Zerner [21, Problem 5.3]) the speed is zero when the expected total drift at a site is at most two and positive above two, with no bounded-stack assumption [22]. About 40 more write-ups of the same kind are on my GitHub. The brain itself is written up in [24].

The perceptron came earlier, in July, when the research brain was running and the math store did not exist yet, and two AI systems did most of the work under my supervision. Krauth and Mézard [25] predicted in 1989 that the largest number M_N of patterns the Ising perceptron stores at zero margin satisfies $M_N/N \to \alpha_\star \approx 0.833$. Ding and Sun [26] proved the lower bound and Huang [27] the upper bound, each conditional on a numerical condition about a low-dimensional variational problem that had been supported only by floating-point computation. My paper [28] verifies both outstanding conditions in certified interval arithmetic and re-derives the parameter rectangle, with $\alpha_\star \in [0.833078599,\ 0.833078600]$. This closes the conditional reduction, for Gaussian disorder and, by universality [29], for subgaussian entries such as ± 1. Huang's condition is a statement over the whole plane with a degenerate point. Reparametrizing by the moments of $(X, \tanh X)$ compresses the plane onto a compact convex body, and near that point, where no box enclosure can work, the proof certifies concavity of a majorant along rays. The certificate layer has a Lean 4 check in the repository [30]. The proof has passed a second-system audit and has not yet been independently reproduced.

I want two theorems. The first answers the question section 3 left open, when links add anything the words do not. Take a store of N cards with K planted topics, in which a card's words and links each reveal its topic with fixed strengths, and let ρ be the correlation of the two channels' errors given the topic. My conjecture is that as N grows there is a threshold ρ^*, depending on the strengths, below which a walk over

the links strictly raises the fraction of held-out questions answered in the first k, and above which that gain vanishes. The word overlap in section 3 suggests my stores may be on the upper side, but I have not estimated ρ. The test is to lower the correlation in a held-out copy and see whether a gain appears. The second is about loops. Model an agent as a Markov chain on routes, where a dead route is absorbing and a record of share f of the known dead routes removes them from the agent's transitions, count a step as useful only if it goes through where the best known proof failed or proves the statement on a new part of its range, and work out how the expected number of useful steps per budget depends on f.

6. The next year

A year from now agents should run for much longer before they need me, on larger and harder problems, connecting areas of research that were not known to be connected. Recall has to get ten times faster and more exact. Section 3 took the easy part, not starting over on each call, and the rest has to come from mathematics. A floor on activation saved little on the separate walk that adds associations to a recall result and changed answers, so pruning that walk needs a bound on what it can change at the top of the list. Personalized PageRank has such a bound (the push algorithm of Andersen, Chung and Lang [31] keeps each entry within ε times its degree at a cost of order $1/(\varepsilon\alpha)$ for a seed of mass one, with α the restart probability, whatever N is), so a z-scored seed must first be clipped or split into positive and negative parts. The weight 0.3 came from a sweep on a development bank and the decay d was set by hand. Both can be fitted by maximum likelihood on a bank where we know the answer cards. The target is a tenth of the median latency on the same bank, with accuracy held within a point.

The bigger job is agency. An agent still chooses its next route by reading, and I want that choice to be a decision rule whose regret I can bound. Treat it as a bandit, where each route r has an unknown probability p_r of yielding a useful step and Thompson sampling [32] over Beta posteriors gives the order in which agents try routes. The record then becomes a posterior shared across agents and days. A route is dropped when the posterior probability that p_r exceeds a floor falls below a fixed level. The test the frontier layer needs is the same open problems with the record and without it, with useful steps per budget as the statistic (a blind test of results on 13 problems [1] was too small to tell). All my tests so far use one group's record, so I also want to repeat them on another group's, with its own agents and no shared history.

Which properties of these graphs predict that a brain will work? On the largest connected part of each graph, at a fixed time, P is a Markov chain, and I want a mixing quantity, the spectral gap or something local around the seed, that predicts the depth at which recall stops improving. Different types of centralities could choose what to serve. The topological and community structure of the mistakes may show where agents fail together. And do learned edges find connections nobody knew? Split the record by date and ask whether an edge between two fields learned before the split predicts a typed link or a result written after it. If it does, agents can follow those edges to new problems.

7. Random matrices and the stability of deep networks

The agents work on my other mathematics from this same memory. That mathematics started with my thesis under Leonid Berlyand, on which part of a trained network is signal and which is noise. Write a trained layer, an $N \times M$ weight matrix, as $W = S + R$, with S of low rank and R noise with independent mean-zero entries of variance σ^2/N and finite fourth moments. The Marchenko-Pastur law [33] says that as N grows with $c = M/N \le 1$ fixed, the singular values of R settle on a fixed density on $[\sigma(1-\sqrt{c}), \sigma(1+\sqrt{c})]$, so for a square matrix with $\sigma = 1$ a singular value at 3 is not noise. Berlyand, Sandier, Zhang and I [34] proved that when a layer has this form, with the singular values of S above the edge of this density (the bulk), for absolute-value activations, removing the singular values below that edge keeps the classification. With high probability an object classified with confidence above an explicit threshold is still classified correctly after the cut, which bounds the pruned accuracy from below. In our experiments on fully connected networks the cut increases accuracy. With Berlyand, Bourdais and Owhadi I carried this method to vision transformers and modern CNNs, using a regularizer that pushes the bulk and the outlying singular values (spikes) apart during training [35, 36]. My preprint on rectangular spikes treats the mesoscopic case, where the signal rank grows with N, and gives the outlier values and singular subspaces [37].

I also study whether a random network is stable at initialization. Take a square ReLU network of width n and depth L with independent $N(0, \alpha/n)$ weights, and let K be its global Euclidean Lipschitz constant, the largest operator norm of a Jacobian on a nonempty activation region. The answer is a transition at $\alpha = 2$, the value He initialization [38] uses. For each $\alpha < 2$ a sufficiently large fixed depth gives $K \le 1$ with exponentially high probability as $n \to \infty$, and for $\alpha > 2$ every fixed depth is expansive. For depth L let α_L be the variance below which that depth gives $K \le 1$. Then $\alpha_L \to 2$, with the certified lower bound $2\exp[-(\sqrt{10} + o(1))\sqrt{\log L/L}]$ [39]. The difficulty is that the region and its Jacobian depend on the same weights. I handle this with coordinate sign changes that preserve the Gaussian law, together with Cover's hyperplane count [40], which cuts the entropy of the activation patterns from $nL\log 2$ to order $n\log L$. A recurrent network reuses one Gaussian W of entry variance $1/N$ at every step, $x_{t+1} = \mathrm{ReLU}(Wx_t)$, and the layers are no longer independent. There I prove exponentially concentrated norm propagation, $\|x_t\|^2 \to 2^{-t}$, over every fixed time horizon for unit starts independent of W [41]. In the ordered phase, where the squared size of a small perturbation shrinks by a factor $\chi_1 < 1$ per layer, I have the exact constant for the smallest eigenvalue of the neural tangent kernel [42] of a deep tanh network at infinite width, for n distinct inputs on the fixed-point sphere, which sets the speed of gradient descent in the kernel regime: $\lambda_{\min} \sim \kappa\lambda_\perp L\chi_1^L$ [43]. I prove that the input-dependent factor $\lambda_\perp$ is positive, so the order Xiao, Pennington and Schoenholz [44] found for the non-constant eigenvalues is attained by the smallest one. With Berlyand, Sarapin, Slavin and Sodin [45] I studied the fixed points and the edge of chaos of randomly initialized autoencoders.

A few others. Under a uniform doubling condition on the data measure, accuracy cannot collapse while the cross-entropy keeps falling [46]. For random data with noisy labels, any two-layer network with m piecewise-linear neurons and arbitrary weights that fits below the noise level has Lipschitz constant at least of order $\sqrt{n/m}$ on n points, up to one logarithmic factor, which is the law of robustness conjectured by Bubeck, Li and Nagaraj [47]; see [48]. The smallest least singular value over all m-row submatrices of a tall Gaussian matrix has an exact exponential scale, and at the critical $N = 2m - 1$ real measurements this gives the sharp base $1/4$ in the Balan-Wang stability parameter [49] of phase retrieval [50]. With Werner Krauth (ENS Paris, manuscript not yet released [51]) I work on a diffusion sampler with no learned score, whose backward path is a mixture of Gaussian bridges to p sample points. Driven by the same noise, the paths $X^{(p)}$ and $X^{(\infty)}$ (from the true target) differ, on times away from the endpoint and for sub-Gaussian targets, by order $1/\sqrt{p}$, with an explicit Gaussian limit.

8. Kernels, and flows on the Bitcoin graph

At NASA's Jet Propulsion Laboratory in 2024 I worked on neural emulators of radiative transfer for imaging spectroscopy and on low-rank and sparse kernel methods. My later paper [52] uses a network g_θ for the mean plus a kernel regression on its residuals: $f(x) = g_\theta(x) + k(x, X)(k(X, X) + \lambda I)^{-1}(y - g_\theta(X))$. On public data, a Matérn kernel on the residuals and on learned features matches the best published result on structural mechanics [53] and improves on the published Gaussian-process emulator for OCO-2 [54] on its own test points, outright on two of three spectral bands. Next I want the agents working on better kernel methods, better kernels with DNNs, and improvements in regression, classification, pruning and adversarial robustness. The theorems of section 7 go in as linked cards, and the idea is to improve all of it while the brain is improving.

Bitcoin tracing is useful for following dirty money and memories are useful for building more agentic AI. Flows through Bitcoin and flows through memories have connections, which is why I study both. The question on the chain is what share of the value at a sink came from a given source, across tens of hops. On 96 real blocks my one-pass tracer handled all 161,524 sources with about 5,500 times less merge work than one trace per source, and 12 sources traced again separately matched to the satoshi. We have not traced a real criminal case. Value is conserved at each transaction, and with a nonnegative seed and damping for cycles personalized PageRank conserves mass too, so a card's score becomes the share of the question's mass that stops there, the same kind of quantity as a source's share at a sink.

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